Chapter 12: Ordinary Differential Equations — Complete Solutions

1. Prove that the following function is a solution of the corresponding ordinary differential equation: $y = e^x$, $e^{-x}\frac{\mathrm{d}y}{\mathrm{d}x} + y^2 - 2ye^x = 1 - e^{2x}$

Proof:

Since $y = e^x$, differentiating gives $\frac{\mathrm{d}y}{\mathrm{d}x} = e^x$.

Substituting y and $\frac{\mathrm{d}y}{\mathrm{d}x}$ into the left-hand side (LHS) of the differential equation:

$$\mathrm{LHS} = e^{-x} \cdot (e^x) + (e^x)^2 - 2(e^x) \cdot e^x = 1 + e^{2x} - 2e^{2x} = 1 - e^{2x}$$

The left-hand side equals the right-hand side (RHS), so $y = e^x$ is a solution of this ordinary differential equation.

2. Find the general solutions of the following ordinary differential equations

(1) $\frac{\mathrm{d}y}{\mathrm{d}x} = x(2y + 1)$

Solution: A separable differential equation.

$$\frac{\mathrm{d}y}{2y+1} = x\mathrm{d}x \implies \frac{1}{2}\ln|2y+1| = \frac{1}{2}x^2 + C_1 \implies \ln|2y+1| = x^2 + C_2$$

$$2y + 1 = Ce^{x^2} \implies y = Ce^{x^2} - \frac{1}{2}$$

(2) $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y}{x} + \sqrt{\left(\frac{y}{x}\right)^2 + 1}$

Solution: A homogeneous differential equation. Let $y = ux \implies \frac{\mathrm{d}y}{\mathrm{d}x} = u + x\frac{\mathrm{d}u}{\mathrm{d}x}$.

$$u + x\frac{\mathrm{d}u}{\mathrm{d}x} = u + \sqrt{u^2+1} \implies x\frac{\mathrm{d}u}{\mathrm{d}x} = \sqrt{u^2+1} \implies \frac{\mathrm{d}u}{\sqrt{u^2+1}} = \frac{\mathrm{d}x}{x}$$

Integrating both sides gives: $\ln(u + \sqrt{u^2+1}) = \ln|x| + \ln C \implies u + \sqrt{u^2+1} = Cx$.

Substituting back $u = \frac{y}{x}$: $\frac{y}{x} + \sqrt{\frac{y^2}{x^2}+1} = Cx \implies y + \sqrt{x^2+y^2} = Cx^2$.

(3) $\frac{\mathrm{d}y}{\mathrm{d}x} = 3y - 2$

Solution:

$$\frac{\mathrm{d}y}{3y-2} = \mathrm{d}x \implies \frac{1}{3}\ln|3y-2| = x + C_1 \implies 3y - 2 = Ce^{3x} \implies y = Ce^{3x} + \frac{2}{3}$$

(4) $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3y}{x} + \frac{x+1}{x}$

Solution: A standard first-order linear nonhomogeneous equation: $\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{3}{x}y = 1 + \frac{1}{x}$.

The integrating factor is $I(x) = e^{\int -\frac{3}{x}\mathrm{d}x} = x^{-3}$. Multiplying both sides by x^{-3}:

$$\frac{\mathrm{d}}{\mathrm{d}x}(yx^{-3}) = (1 + \frac{1}{x})x^{-3} = x^{-3} + x^{-4}$$

$$yx^{-3} = \int (x^{-3} + x^{-4})\mathrm{d}x = -\frac{1}{2}x^{-2} - \frac{1}{3}x^{-3} + C \implies y = -\frac{1}{2}x - \frac{1}{3} + Cx^3$$

(5) $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2y}{x+1} + (x+1)^3$

Solution: Rewrite as $\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{2}{x+1}y = (x+1)^3$.

The integrating factor is $I(x) = e^{\int -\frac{2}{x+1}\mathrm{d}x} = (x+1)^{-2}$. Multiplying both sides by $(x+1)^{-2}$:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[y(x+1)^{-2}\right] = (x+1)^3 \cdot (x+1)^{-2} = x+1$$

$$y(x+1)^{-2} = \int (x+1)\mathrm{d}x = \frac{1}{2}(x+1)^2 + C \implies y = \frac{1}{2}(x+1)^4 + C(x+1)^2$$

3. Find the general solutions of the following ordinary differential equations

(1) $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{2}{1-x}\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

Solution: A higher-order differential equation that can be reduced in order. Let $p = \frac{\mathrm{d}y}{\mathrm{d}x}$; then $\frac{\mathrm{d}p}{\mathrm{d}x} + \frac{2}{1-x}p = 0$.

$$\frac{\mathrm{d}p}{p} = \frac{2}{x-1}\mathrm{d}x \implies \ln|p| = 2\ln|x-1| + \ln C_1 \implies p = C_1(x-1)^2$$

Since $p = \frac{\mathrm{d}y}{\mathrm{d}x}$, integrate once more:

$$y = \int C_1(x-1)^2\mathrm{d}x = C_1(x-1)^3 + C_2$$

(2) $2\frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 1$

Solution: Let $p = \frac{\mathrm{d}y}{\mathrm{d}x}$; the equation becomes $2p\frac{\mathrm{d}p}{\mathrm{d}x} = 1 \implies 2p\mathrm{d}p = \mathrm{d}x$.

Integrating both sides gives: $p^2 = x + C_1 \implies p = \pm\sqrt{x + C_1}$.

That is, $\frac{\mathrm{d}y}{\mathrm{d}x} = \pm\sqrt{x + C_1}$; integrating once more gives the general solution:

$$y = \pm\frac{2}{3}(x + C_1)^{\frac{3}{2}} + C_2$$

4. Prove that $y = (c_1 + c_2x)e^{-2x} + \frac{1}{8}\sin 2x$ is the general solution of the equation $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = \cos 2x$

Proof:

1. **Verification of the general solution of the homogeneous equation**: the characteristic equation is $r^2 + 4r + 4 = 0 \implies (r+2)^2 = 0$, with repeated roots $r_{1,2} = -2$.

   The corresponding general solution of the homogeneous equation is $y_h = (c_1 + c_2x)e^{-2x}$.

2. **Verification of the particular solution of the nonhomogeneous equation**: let the particular solution be $y_p = \frac{1}{8}\sin 2x$. Differentiating:

   $$y_p' = \frac{1}{4}\cos 2x, \quad y_p'' = -\frac{1}{2}\sin 2x$$

   Substituting into the left-hand side of the nonhomogeneous equation:

   $$\mathrm{LHS} = -\frac{1}{2}\sin 2x + 4\left(\frac{1}{4}\cos 2x\right) + 4\left(\frac{1}{8}\sin 2x\right) = \cos 2x = \mathrm{RHS}$$

   It satisfies the structure $y = y_h + y_p$ and contains two independent arbitrary constants, which completes the proof.

5. Prove that $y = c_1 \cos x + c_2 \sin x - \frac{1}{3} x \cos 2x + \frac{4}{9} \sin 2x$ is the general solution of the equation $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + y = x \cos 2x$

Proof:

1. **Homogeneous part**: the characteristic equation is $r^2 + 1 = 0 \implies r = \pm i$, and the general solution of the homogeneous equation is $y_h = c_1 \cos x + c_2 \sin x$.
2. **Particular solution of the nonhomogeneous equation**: let the particular solution be $y_p = -\frac{1}{3} x \cos 2x + \frac{4}{9} \sin 2x$. Differentiating:

$$y_p' = -\frac{1}{3} \cos 2x + \frac{2}{3} x \sin 2x + \frac{8}{9} \cos 2x = \frac{5}{9} \cos 2x + \frac{2}{3} x \sin 2x$$

$$y_p'' = -\frac{10}{9} \sin 2x + \frac{2}{3} \sin 2x + \frac{4}{3} x \cos 2x = -\frac{4}{9} \sin 2x + \frac{4}{3} x \cos 2x$$

Substituting into the left-hand side of the equation:

$$\mathrm{LHS} = y_p'' + y_p = \left(-\frac{4}{9} \sin 2x + \frac{4}{3} x \cos 2x \right) + \left(-\frac{1}{3} x \cos 2x + \frac{4}{9} \sin 2x \right) = x \cos 2x = \mathrm{RHS}$$

Since it contains two arbitrary constants and has the complete form, this expression is indeed the general solution of the equation.

6. Find the general solutions of the following ordinary differential equations

(1) $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$

Solution: The characteristic equation is $r^2 - 3r + 2 = 0 \implies (r-1)(r-2) = 0 \implies r_1 = 1, r_2 = 2$.

The general solution is: $y = C_1 e^x + C_2 e^{2x}$.

(2) $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3y = 0$

Solution: The characteristic equation is $r^2 - 3 = 0 \implies r = \pm\sqrt{3}$.

The general solution is: $y = C_1 e^{\sqrt{3}x} + C_2 e^{-\sqrt{3}x}$.

(3) $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 5\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = x$

Solution:

1. **General solution of the homogeneous equation**: the characteristic equation is $r^2 + 5r + 4 = 0 \implies r_1 = -1, r_2 = -4 \implies y_h = C_1 e^{-x} + C_2 e^{-4x}$.
2. **Particular solution of the nonhomogeneous equation**: let the particular solution be $y_p = Ax + B \implies y_p' = A, y_p'' = 0$.

   Substituting into the equation: $5A + 4(Ax + B) = x \implies 4Ax + (5A + 4B) = x$.

   Equating corresponding coefficients: $4A = 1 \implies A = \frac{1}{4}$; $5A + 4B = 0 \implies B = -\frac{5}{16}$.

   The general solution is: $y = C_1 e^{-x} + C_2 e^{-4x} + \frac{1}{4} x - \frac{5}{16}$.

(4) $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + y = 5e^{2x}$

Solution:

1. **General solution of the homogeneous equation**: the characteristic equation is $r^2 + 1 = 0 \implies r = \pm i \implies y_h = C_1 \cos x + C_2 \sin x$.

2. **Particular solution of the nonhomogeneous equation**: let the particular solution be $y_p = Ae^{2x} \implies y_p' = 2Ae^{2x}, y_p'' = 4Ae^{2x}$.

   Substituting into the equation: $4Ae^{2x} + Ae^{2x} = 5e^{2x} \implies 5A = 5 \implies A = 1$.

   The general solution is: $y = C_1 \cos x + C_2 \sin x + e^{2x}$.